

Growth and Development



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Growth and Development

- **Growth:** It means increase in the size of the body as a whole or its individual parts.
- **Development:** means maturation of systems and organs as well as achievement of skills and abilities whether motor, mental, emotional or social.



Growth and Development

❑ **Factors affecting growth and development:**

1- Genetic factors.

2- Nutritional: decreased proteins, minerals or vitamins quantitatively or qualitatively may cause retarded growth and development. Debilitating diseases interfere with adequate nutrition e.g. chronic diarrhea, parasitic infestation and tuberculosis. Over nutrition may cause obesity.

3- Socio-economic: poverty is associated with poor growth.





Growth and Development

❑ **Factors affecting growth and development:**

4- Chronic diseases: if starting early in life may cause retarded growth e.g. chronic asthma, chronic renal failure and liver cirrhosis.

5- Endocrinal disorders: e.g. hypothyroidism.

6- Intrauterine infections: e.g. congenital rubella

7- Exposure to environmental pollutants: e.g lead and mercury.

8- Emotional deprivation: Emotional deprivation can lead to impaired growth and development.

Growth and Development

❑ Growth Measurements

- Weight.
- Height (length).
- Head Circumference.
- Chest Circumference.
- Mid arm Circumference.



Growth and Development

❖ **Assessment of growth**

❑ **Anthropometric measurements**

(A) Weight

- In the first 4 months of life, the weight increases by ¾ Kg/month, next 4 months the weight increases ½ Kg/month and in the last 4 months, it increases by about ¼ Kg/month. The infant doubles his birth weight by 5 months and triples it at the end of the first year.
- During the preschool period (2-5yr), the child gains about 2 Kg/year.
- During the early school years, the child gains about 3-3.5 Kg/year.

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- ❖ **Assessment of growth**
- ❑ **Anthropometric measurements**

(A) Weight







Growth and Development

❖ **Assessment of growth**

❑ **Anthropometric measurements**

(A) Weight

Age	**Average weight (Kg)**
At birth	3.25
3-12 mo	(age (mo) + 9) / 2
1-6 years	age (yr) x 2 + 8
7-12years	[(age (yr) x 7) – 5] / 2

Growth and Development

❖ **Assessment of growth**

❑ **Anthropometric measurements**

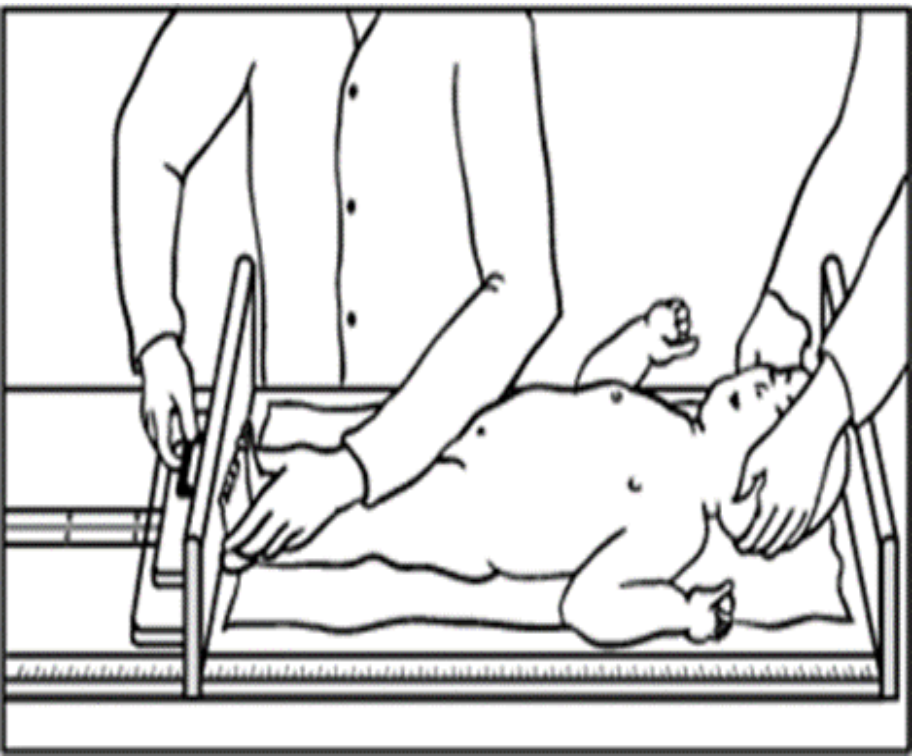
(B) Height (length)

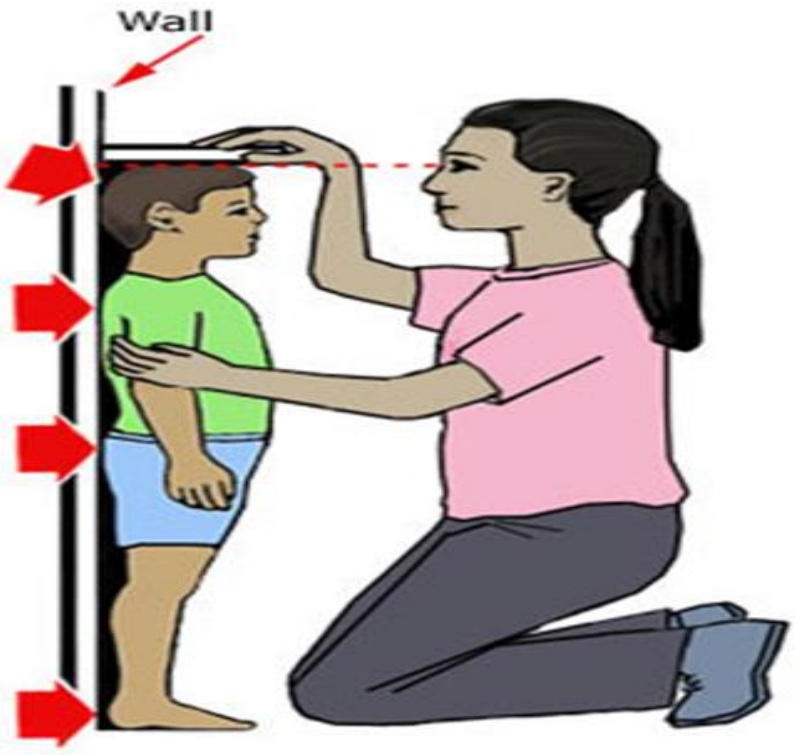
Age	**Average (cm)**
At birth	50
At year	75
At 2 years	87.5
3-12 years	(age (yr)x6) + 77 = …. Or (age (yr) x 5) + 80 = ….

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- ❖ **Assessment of growth**
- ❑ **Anthropometric measurements**

(B) Height (length)





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❖ **Assessment of growth**

❑ **Anthropometric measurements**

(C) Head circumference (HC):

Age	**Average HC (cm)**
At birth	35
At 6 mo	44
At 1 yr	47
At 2 yr	49

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❖ **Assessment of growth**

❑ **Anthropometric measurements**

(C) Head circumference (HC):



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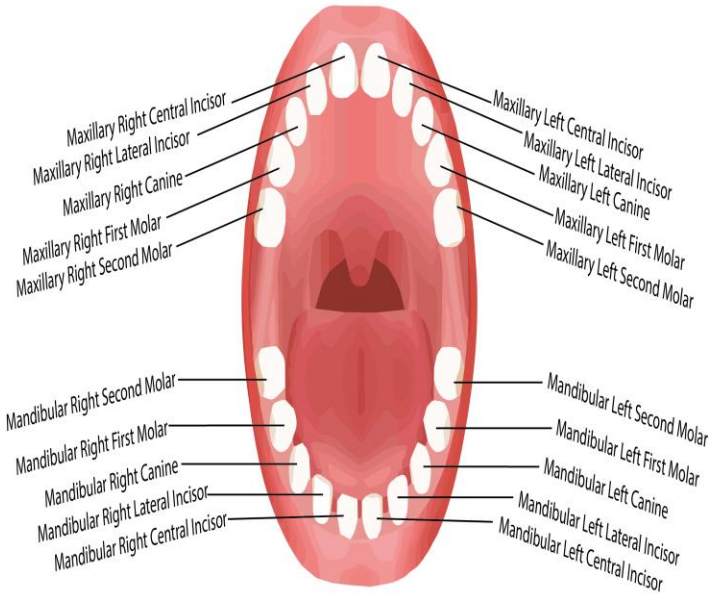
❖ **Assessment of growth**

❑ **Anthropometric measurements**

(D) Tooth eruption

- Tooth eruption may be delayed up to one year of age.

Deciduous (primary)	**Average age of eruption**
Central incisors	6th month
Lateral incisors	9th month
First molars	12th month
Canines	18th month
Second molars	24th month

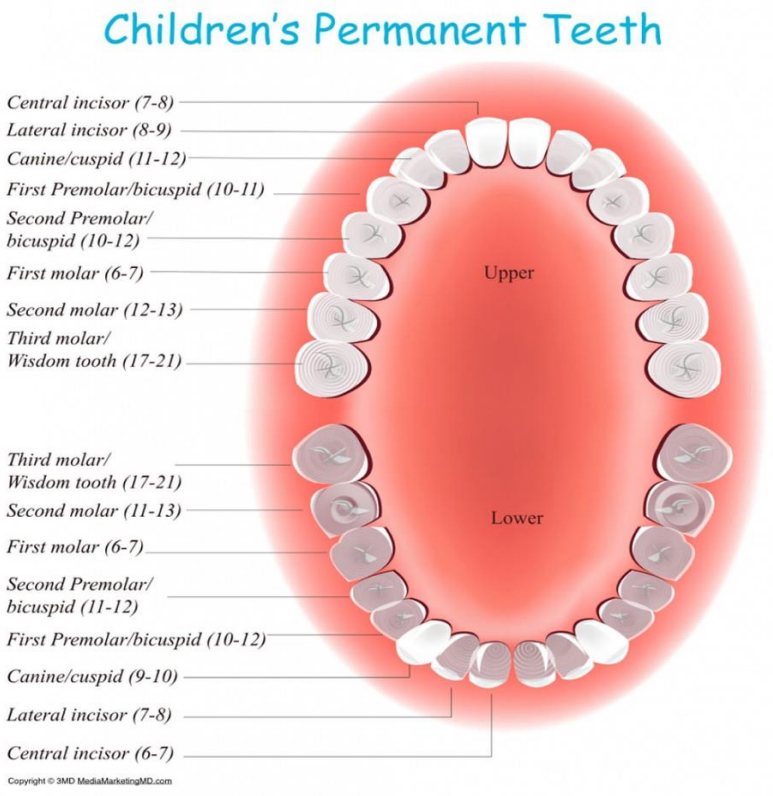


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- **Assessment of growth**
  - **Anthropometric measurements**

(D) Tooth eruption

Permanent (Secondary)	Average age of eruption
First molar	6th year
Central incisors	6th year
Lateral incisors	7th year
Canines	10th year
First premolars	10th year
Second premolars	11th year
Second molars	11th to 12th year
Third molar	20th year



Growth and Development

❑ **Motor development during infancy**

At birth: lies in flexed attitude and turns head from side to side.

3 mo: there is some head control (head support).

5 mo: sits with trunkal support and takes objects to the mouth.

6 mo: sits without support.

9 mo: crawls.

10 mo: stands with support.

12mo: walks, with one hand held.





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- **Mental and social development during infancy**



4 wk: watches persons or bright object.

2mo: the infant smiles on social contact, listens to voice.

3mo: says (aah, ngah).

4mo: Laughs, may show displeasure if social contact is broken, excited at sight of food.

7mo: prefers mother.

10 mo: repetitive consonant sounds (mama, dada) responds to sound of name and waves bye-bye.

12mo: a few words besides "mama, dada" He plays simple ball game and makes postural adjustment to dressing.

Growth and Development

- **Causes of delayed motor development without mental retardation**

1. Familial and emotional disturbances.
2. Mild protein energy malnutrition.
3. Rickets.
4. Poliomyelitis.
5. Chronic diseases.
6. Muscle, bone and joint diseases.



Growth and Development

- **Causes of delayed motor development with mental retardation**



1. Cerebral palsy.

2. Chromosomal abnormality e.g. down syndrome.

3. Hypothyroidism.

4. Inborn error of metabolism.

5. CNS infection.

6. Intrauterine infections.






Thank You!